

From Mathematical Foundations to Regulatory Practice

Quantitative Methods Behind Model Risk, AI Governance, and Machine Learning in Financial Services

How four foundational techniques — stochastic calculus, non-convex optimization, Fourier analysis, and wavelet transforms — underpin derivatives valuation, model validation, ML development, and time-series analysis, and where each connects to current model-risk and AI-governance frameworks (SR 26-2, NIST AI RMF, the EU AI Act).

This is an expository piece: it connects four well-established mathematical methods — Fourier analysis, stochastic (Itô) calculus, non-convex optimization, and wavelet transforms — to concrete problems in regulated financial services. The aim is to make the link between the underlying mathematics and day-to-day model-risk and AI-governance work explicit and legible. Each method is presented with a worked example and a short note on its governance implications. Full mathematical derivations of every result are available in a companion technical document.

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A note on scope and authorship. The mathematical results here are classical and long-established; the contribution of this document is the synthesis — mapping foundational methods onto regulated-industry applications in a single, transparent reference. It was prepared as a rigorous exposition, with all computations independently verified, and is intended to demonstrate the connection between quantitative fundamentals and applied model-risk and AI-governance practice.

1 · Stochastic Calculus — the Engine of Derivatives Pricing and Model Validation

Almost every quantitative model that touches a price over time — option valuation, interest-rate models, credit exposure, counterparty risk — is built on stochastic calculus. The central object is the Itô integral, which makes sense of integrating against a random, jagged price path where ordinary calculus fails.

The defining feature is the **Itô correction**. For Brownian motion B , the integral of the process against itself is

$$\int_0^T B_t dB_t = \frac{1}{2}(B(T)^2 - T),$$

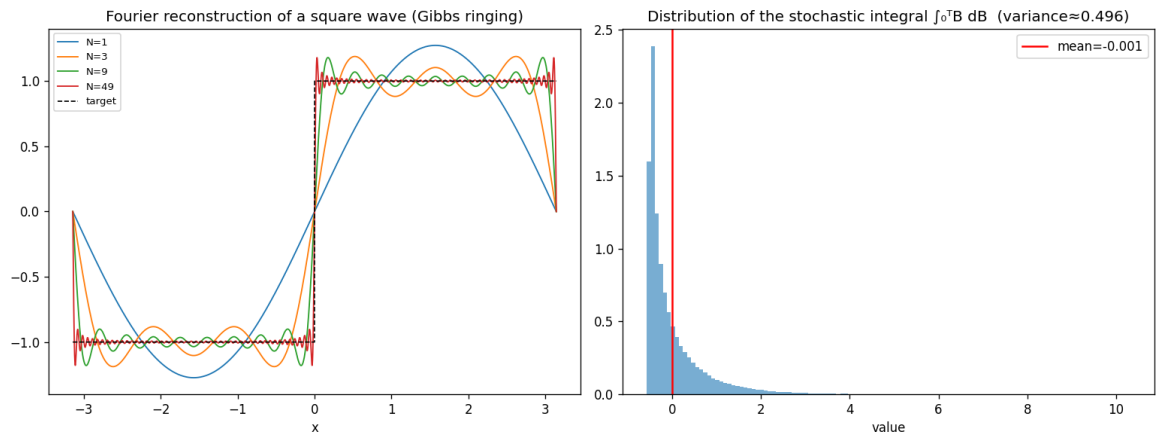
not the $\frac{1}{2}B(T)^2$ that ordinary calculus would suggest. The extra $-\frac{1}{2}T$ term arises because a random path accumulates ‘quadratic variation’ that a smooth curve does not. The same correction, applied to the logarithm of a stock price, produces the $-\sigma^2/2$ drift in the geometric-Brownian-motion model underlying Black-Scholes:

$$S(T) = S_0 \exp((r - \sigma^2/2)T + \sigma B(T)), \text{ Call} = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2).$$

For a representative call ($S_0=100$, $K=105$, $r=3\%$, $\sigma=20\%$, $T=1$), the closed-form price is 7.13, which a Monte-Carlo simulation over two million paths reproduces to within a cent — a basic example of the analytic-vs-numerical cross-check that validation work depends on.

Why it matters in practice. Pricing models, value-at-risk engines, and exposure calculators are stochastic-calculus models under the hood. Anyone validating them needs to recognize where the Itô correction lives — dropping or mis-implementing that $-\sigma^2/2$ drift is a real and recurring source of model error.

Governance & model-risk lens. U.S. model-risk supervision was substantially revised in 2026. On April 17, 2026 the Federal Reserve, OCC, and FDIC issued *Revised Guidance on Model Risk Management* (Federal Reserve SR 26-2), which superseded the long-standing SR 11-7 (2011) and moved to a more principles-based, risk-tiered framework.^[5,6] The three core validation pillars — conceptual soundness, outcomes analysis, and ongoing monitoring — are retained, though the revised guidance is more proportionate and de-emphasizes prescriptive organizational independence. The analytic-vs-simulation benchmark above is precisely the kind of independent check that supports conceptual soundness before a pricing model is approved for use. (Guidance current as of June 2026; supervisory expectations continue to evolve.)



Right panel: the simulated distribution of the stochastic integral $\int_0^T B \, dB$, centered at zero with variance $T^2/2$ — the kind of Monte-Carlo output a model-validation team reconciles against a closed-form result. (Left panel relates to Section 3.)

2 · Non-Convex Optimization — the Core of Machine-Learning Model Training

Training a machine-learning model means minimizing a loss function over its parameters. For anything beyond the simplest linear models, that loss surface is **non-convex** — it has many valleys (local minima), not one. Gradient-based training finds a valley, but not necessarily the deepest one.

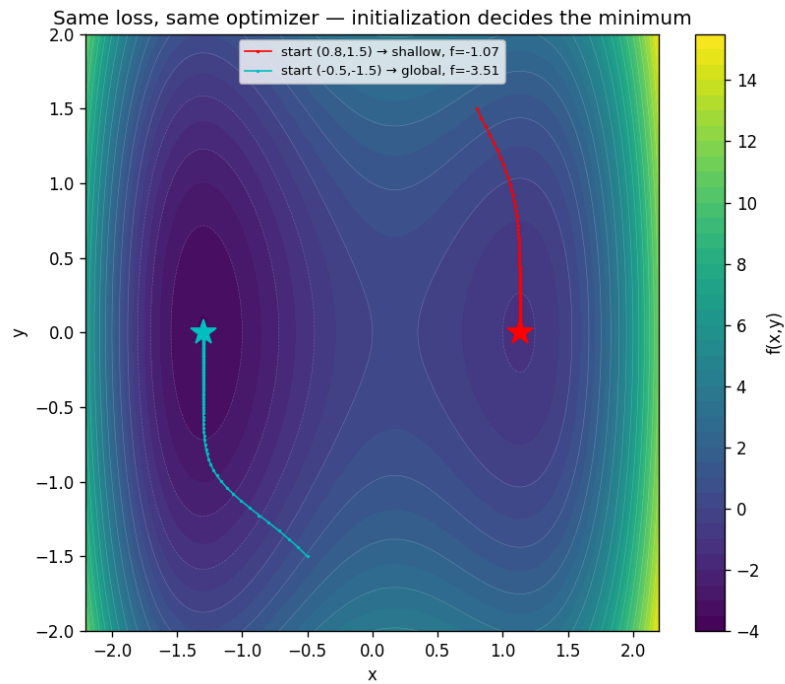
A one-line illustration: the function $f(x) = x^4 - 3x^2 + x$ has two local minima — a shallow one at $x \approx 1.13$ (value -1.07) and the true global one at $x \approx -1.30$ (value -3.51) — separated by a ridge. A descent algorithm started on the wrong side of that ridge converges to the shallow minimum and never reaches the global one. Extended to two dimensions, the same optimizer reaches different answers purely depending on where it starts.

***Why it matters in practice.** This is why training a model twice can give two different models. The outcome depends on initialization, the order data is seen, and the optimizer's settings. It is not a bug — it is the geometry of non-convex loss. Every modern deep-learning model is the product of this process.*

Governance & model-risk lens. For model risk and AI governance this has direct consequences: (1) **Reproducibility** — results must be controlled with fixed random seeds and documented training configurations, or the model is not reproducible for audit. (2)

Stability — a model that lands in a different minimum on retraining may behave differently on the same inputs, which validation and ongoing monitoring must detect. (3)

Documentation & governance — the NIST AI Risk Management Framework (AI RMF 1.0, 2023) organizes this work into four functions — Govern, Map, Measure, and Manage — and expects the development process, including these non-determinism controls, to be recorded and reviewable.^[7] The 2026 model-risk guidance (SR 26-2) likewise emphasizes lifecycle governance and continuous monitoring.^[5] Note a current scope boundary: SR 26-2 treats generative and agentic AI as outside its direct scope, signaling that those systems are likely to be addressed through separate, still-developing guidance.



Two runs of the same optimizer on the same loss surface. Initialization alone decides whether it reaches the global minimum (cyan) or is trapped in a shallow local minimum (red) — a concrete picture of why ML training is non-deterministic and must be governed accordingly.

3 · Fourier Analysis — Frequency Structure in Time Series and the Limits of Approximation

Fourier analysis decomposes a signal — a price series, a volatility surface, a stream of transactions — into its constituent frequencies. It is a standard tool for detecting cycles and seasonality and for filtering noise in time-series and signal-processing models.

Its most instructive limitation is the **Gibbs phenomenon**: when a signal has a sudden jump, a finite sum of smooth waves cannot reproduce it cleanly. The approximation overshoots the jump by a fixed amount — about 9% of the jump height — and this overshoot does *not* disappear as more terms are added; it only narrows.

***Why it matters in practice.** Any model that reconstructs or smooths a signal from a finite set of frequency components will ‘ring’ near sudden changes — a market shock, a regime break, a structural shift. That ringing is an artifact of the method, not a feature of the data. Mistaking it for signal leads to spurious indicators and false alerts.*

Governance & model-risk lens. A validator should know whether a model's apparent oscillations near discontinuities are real or are Gibbs artifacts of a Fourier-based transform. Misreading the artifact as predictive signal is precisely the kind of conceptual-soundness flaw independent review is meant to catch before a model influences decisions.

The left panel of the figure in Section 1 shows this directly: as the number of terms grows (1, 3, 9, 49), the reconstruction of a square wave sharpens but the spike at the jump persists at the same height.

4 · Wavelet Transforms — Localized, Multiscale Features for Signals and Risk Data

Where Fourier analysis tells you *which* frequencies are present, wavelet analysis also tells you *where* and *at what scale*. It decomposes a signal into localized bumps at multiple resolutions — capturing both a slow trend and a sudden spike in the same framework. This makes wavelets well suited to financial data, which is full of features that are both localized in time and varied in scale.

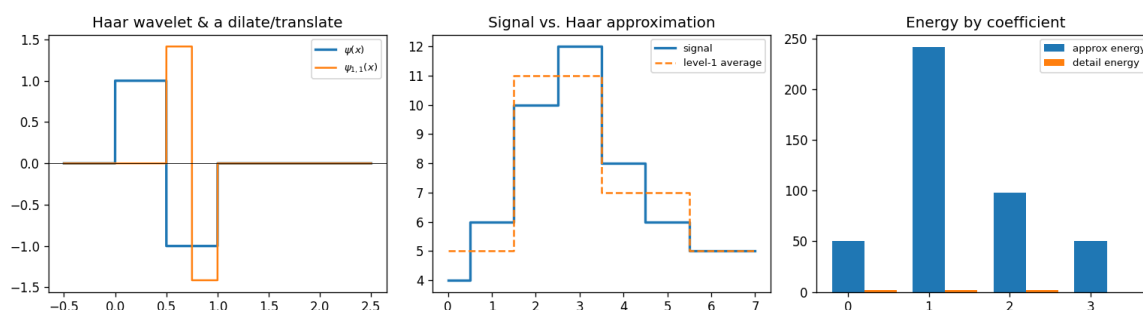
The simplest version, the Haar transform, recodes a signal into **averages** (coarse trend) and **details** (fine changes). For a signal of consecutive pairs, each pair (s_1, s_2) becomes

$$\text{average } a = (s_1 + s_2)/\sqrt{2}, \text{ detail } d = (s_1 - s_2)/\sqrt{2}.$$

This transform is **lossless** (the original is exactly recoverable) and **energy-preserving** ($a^2 + d^2 = s_1^2 + s_2^2$). Where the signal is locally flat, the detail coefficient is zero — so most of the information concentrates in a few coefficients. That sparsity is the basis of wavelet compression and denoising.

***Why it matters in practice.** Wavelet features give models a way to see structure at multiple time scales at once — useful for detecting regime changes, transient anomalies, and multiscale volatility. Because the transform is lossless and invertible, it is a clean, auditable preprocessing step rather than a black box.*

Governance & model-risk lens. For AI governance, the transparency of a wavelet decomposition is an asset: each coefficient has a clear meaning (a specific scale and location), so feature engineering built on it is explainable — a meaningful advantage when a model's inputs must be justified to validators, auditors, or regulators. This matters increasingly under the EU AI Act, which classifies AI used in credit scoring and creditworthiness assessment as *high-risk* and imposes transparency, data-governance, and human-oversight obligations.^[8] (Compliance dates for high-risk systems were deferred in 2026: under the provisionally agreed Digital Omnibus, stand-alone high-risk obligations now apply from 2 December 2027 — current as of June 2026 and subject to formal adoption.)



A Haar wavelet (left), a signal and its coarse approximation (middle), and the concentration of energy into a few coefficients (right) — the property that makes wavelets effective for compression, denoising, and multiscale feature extraction.

Closing

These four methods are foundational, not novel — but the connection drawn here is the point: the same mathematics that underlies derivatives pricing, ML training, and signal analysis is also where much of the model risk lives. Effective model-risk management and AI governance in financial services depend on understanding these foundations well enough to know where each model can fail — the Itô correction that must be implemented correctly, the non-convexity that makes training non-reproducible, the Gibbs ringing that masquerades as signal, the wavelet structure that keeps features explainable. Combining quantitative depth with regulatory and validation expertise is what allows that judgment to be exercised in practice.

Practical takeaway for firms: the strongest defense against enforcement actions and supervisory findings is demonstrable understanding of where each model can fail — implemented controls, documented validation, reproducible training, and explainable features — mapped to the current supervisory frameworks. Combining quantitative depth with regulatory and validation expertise is what makes that defense credible.

A companion technical document provides the complete mathematical derivations and proofs for every result referenced above. All regulatory references in this document are current as of June 2026; supervisory guidance and statutory timelines evolve, and citations should be re-verified against primary sources before reliance.

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- [8] Regulation (EU) 2024/1689 (the EU Artificial Intelligence Act), in force August 1, 2024. Classifies AI used in credit scoring and creditworthiness assessment as high-risk (Annex III). High-risk compliance timelines were amended by the 2026 Digital Omnibus (provisional agreement, May 2026; formal adoption pending), deferring stand-alone high-risk obligations to December 2, 2027.

Note. Regulatory frameworks change. Each citation above reflects the status as of June 2026 to the best of the author’s knowledge; readers should confirm the current text and applicability of any guidance, regulation, or statute against primary sources before relying on it. This document is informational and is not legal, regulatory, or investment advice.